

# Hall Effect and Magnetoresistance Measurement of an Aluminum H-Bar

## Aim

To measure the Hall resistance ( $R_{xy}$ ) and longitudinal magnetoresistance ( $R_{xx}$ ) of an Aluminum (Al) thin film at **300 K** using an applied magnetic field sweep up to **1 T (10,000 Oe)**.

## Apparatus & Setup

Based on standard transport measurement protocols and the physical layout of your experiment:

- **Sample:** Aluminum thin film, patterned into an H-bar geometry via lithography and Thermal Vacuum Deposition (TVD).
- **Mounting:** Resistivity/Hall measurement puck (configured for standard cryostat or PPMS insertion).



Figure : insertion rod assembly

- **Electrical Connections:** Fine wire bonds connecting the H-bar pads to the puck's macroscopic terminal pins.



Figure : insertion rod puck mount

- **Measurement Electronics:** \* AC Current Source (supplying **1 mA**).

- Lock-in Amplifier (for high-precision phase-sensitive detection of micro-volt signals, isolating the fundamental harmonic and rejecting noise).
- **Magnet System:** Electromagnet capable of sweeping fields precisely between **+10,000 Oe** and **-10,000 Oe**.



Figure : DynaCool PPMS (Physical Properties Measurement System)

## **Theory**

When a current  $I$  flows through a conducting strip of thickness  $t$  in the presence of a perpendicular magnetic field  $B$ , the charge carriers experience a Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

This force deflects the carriers to one side of the sample, creating a transverse electric field until the resulting electrostatic force exactly balances the Lorentz force.

### **3.1 The Hall Coefficient and Carrier Concentration**

The accumulation of charge results in a measurable transverse voltage, the Hall Voltage ( $V_H$ ). For a simple single-band metal, the Hall resistance is perfectly linear with respect to the applied magnetic field:

$$R_{xy} = \frac{V_H}{I} = R_H \frac{B}{t}$$

Where  $R_H$  is the Hall coefficient. By measuring the slope of  $R_{xy}$  versus  $B$ , we can extract  $R_H$ . The Hall coefficient is fundamentally linked to the carrier concentration ( $n$ ) and the charge of the carriers ( $q$ ):

$$R_H = \frac{1}{nq}$$

- A negative slope ( $R_H < 0$ ) indicates electron-dominated transport.
- A positive slope ( $R_H > 0$ ) indicates hole-dominated transport.

### 3.2 Parasitic Voltage and Symmetrization

In a physical H-bar device, it is impossible to align the transverse voltage contacts perfectly. Consequently, the measured voltage contains both the true Hall signal and a parasitic longitudinal voltage drop (from the standard resistance of the material):

$$R_{measured}(B) = R_{xy}(B) + R_{xx}(B)$$

Because the Hall effect is an **odd (anti-symmetric)** function of the magnetic field and magnetoresistance is an **even (symmetric)** function, they can be mathematically isolated by sweeping the field in both directions:

- **Hall Resistance:**  $R_{xy} = \frac{R(+B) - R(-B)}{2}$
- **Magnetoresistance:**  $R_{xx} = \frac{R(+B) + R(-B)}{2}$

## Procedure

1. **Sample Preparation:** Following TVD synthesis and lithographic patterning, clean the Aluminum H-bar sample.



Figure : Hall Bar mounted on puck

2. **Mounting & Wire Bonding:** Secure the sample to the measurement puck. Using a wire bonder, make four-terminal connections:
  - Drive a constant current  $I$  along the main longitudinal channel.
  - Connect **Channel 1 (Ch1)** across two opposite arms of the H-bar to measure the transverse Hall voltage ( $V_H$ ).
  - Connect **Channel 2 (Ch2)** across two adjacent arms on the same side to measure the longitudinal voltage ( $V_L$ ) if direct MR measurement is desired.
3. **Initialization:** Insert the puck into the measurement chamber and stabilize the temperature at 300 K.
4. **Measurement Parameters:** Set the lock-in amplifier to supply a 1 mA AC excitation current.
5. **Field Sweeping:** \* Sweep the magnetic field from **+10,000 Oe to -10,000 Oe** (Sweep Down), recording the field, resistance, phase angle, and standard deviation at specified intervals.
  - Sweep the magnetic field back from **-10,000 Oe to +10,000 Oe** (Sweep Up) to check for magnetic hysteresis or instrument drift.
6. **Data Extraction:** Export the data files for computational analysis to separate  $R_{xy}$  and  $R_{xx}$  using the parity formulas described in the theory section.

## Observations

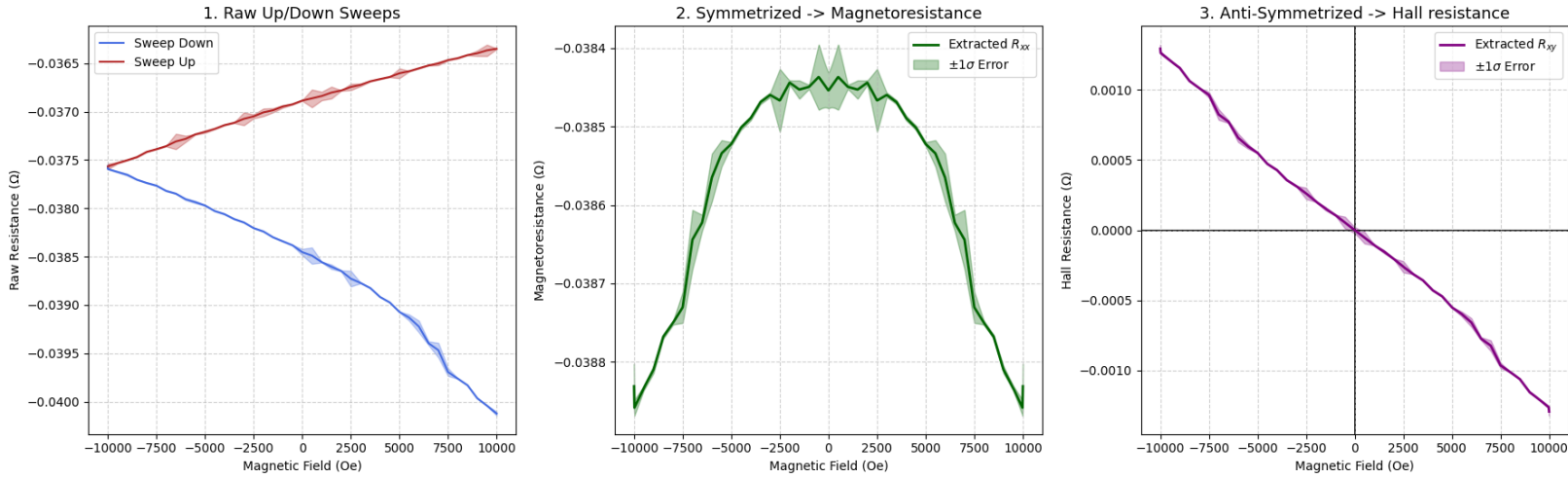


Figure : Plots summarising the observed data

Data in CSV format is available at : <https://github.com/jajapuramshivasai/Advanced-Experimental-Techniques-in-Quantum-Materials/>

## Results

**Average Magnetoresistance:**  $-3.858050 \times 10^{-2} \Omega$

**Net MR Error:**  $5.387036 \times 10^{-7} \Omega$

Note : here Magento resistance proportional to applied field squared so its value is not of main interest

**Extracted Hall Slope ( $R_H$ ):**  $-1.210884 \times 10^{-7} \Omega/\text{Oe}$

**Net Hall Slope Error:**  $2.629448 \times 10^{-10} \Omega/\text{Oe}$

## **Precautions**

- **Thermal Stability:** Ensure the temperature is perfectly stabilized at 300 K before sweeping, as drifting temperatures will alter the base resistivity, ruining the symmetrization math.
- **Phase Angle:** Monitor the phase angle on the lock-in amplifier (maintained near  $180^\circ$  in your data) to ensure the signal is purely resistive and not corrupted by inductive coupling from the wire bonds.
- **Contact Resistance:** While the H-bar geometry inherently suppresses contact resistance issues compared to two-probe methods, high-resistance wire bonds can still increase the thermal noise floor, necessitating larger lock-in time constants.